



Organic Chemistry

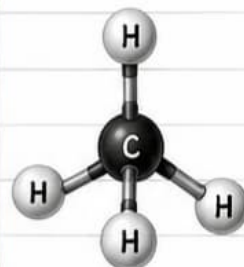
Some Basic Principles and Techniques

1. INTRODUCTION

- Organic chemistry is the study of **carbon** compounds.
- Carbon has unique ability to form **stable covalent bonds** with itself and with other elements.
- This property leads to a very large number of compounds.

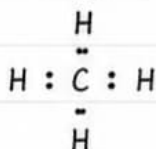
Why Carbon?

- ✓ Tetravalent - forms 4 bonds
- ✓ Catenation - forms long chains, branched chains & rings
- ✓ Strong C-C bond - stable
- ✓ Forms single, double & triple bonds

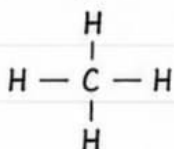
Methane (CH₄)

2. COVALENT BONDING IN ORGANIC COMPOUNDS

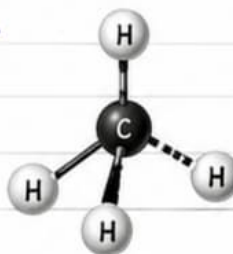
- Carbon forms covalent bonds by **sharing electrons**.



(Electron dot structure of methane)



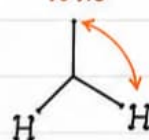
(Line structure of methane)



(3D structure - Tetrahedral)

Bond Angle

109.5°

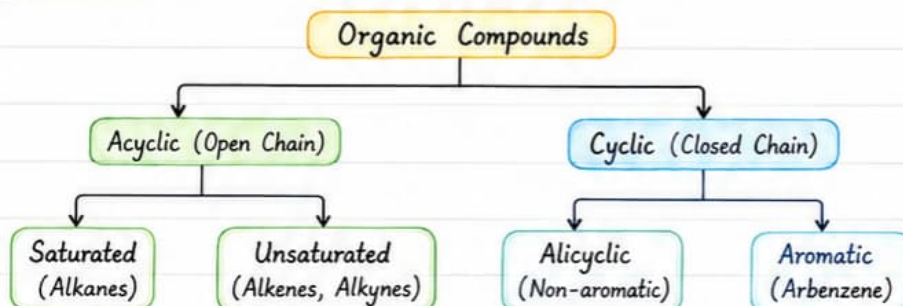


3. HOMOLOGOUS SERIES

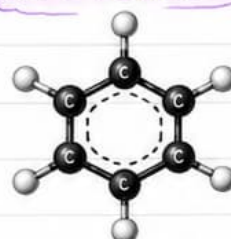
- A family of organic compounds having the **same functional group** and general formula.
- Successive members differ by **-CH₂- unit**.

Series	General Formula	Functional Group	Example	First Member
Alkanes	C_nH_{2n+2}	—	CH ₃ -CH ₃ (Ethane)	CH ₄ (Methane)
Alkenes	C_nH_{2n}	C = C	CH ₂ = CH ₂ (Ethene)	C ₂ H ₄ (Ethene)
Alkynes	C_nH_{2n-2}	C ≡ C	HC ≡ CH (Ethyne)	C ₂ H ₂ (Ethyne)
Alcohols	$C_nH_{2n+1}OH$	-OH	CH ₃ -OH (Methanol)	CH ₃ -OH
Carboxylic Acids	$C_nH_{2n+1}COOH$	-COOH	CH ₃ COOH (Acetic acid)	HCOOH

4. CLASSIFICATION OF ORGANIC COMPOUNDS



Aromatic compounds are more stable due to resonance.

Benzene (C₆H₆)



Organic Chemistry: Some Basic Principles and Techniques

5. NATURE OF CARBON-CARBON BOND

- Carbon atoms can form **strong covalent bonds** with other carbon atoms.
- These bonds can be **single**, **double** or **triple**.
- C-C bonds are very stable due to small size of carbon and effective overlap.
- Hence, long chains, branched chains and rings are possible.

Types of C-C Bonds				
Type	Bond	Hybridisation	Example	3D Model
Single	$C-C$	sp^3	Ethane (CH_3-CH_3)	
Double	$C=C$	sp^2	Ethene ($CH_2=CH_2$)	
Triple	$C\equiv C$	sp	Ethyne ($HC\equiv CH$)	

6. FUNCTIONAL GROUP

- An atom or a group of atoms which determines the characteristic chemical properties of an organic compound is called **functional group**.
- Compounds having the same functional group show similar chemical behaviour.

Some Important Functional Groups			
Functional Group	Representation	Example	Class
$-OH$	$R-OH$	CH_3-OH	Alcohols
$-COOH$	$R-COOH$	CH_3COOH	Carboxylic acids
$-CHO$	$R-CHO$	CH_3CHO	Aldehydes
$\begin{array}{c} \diagup \\ C=O \\ \diagdown \end{array}$	$R-CO-R'$	CH_3COCH_3	Ketones
$-NH_2$	$R-NH_2$	CH_3NH_2	Amines

7. IUPAC NOMENCLATURE - BASIC RULES

- Select the **longest** continuous carbon chain containing the functional group / multiple bond.
- Number the chain from the end **nearer** to the functional group or multiple bond.
- Replace the **suffix 'e'** of parent alkane by appropriate suffix.

Examples

$\begin{array}{ccccccc} & 3 & 2 & 1 & & & \\ & CH_3 & -CH_2- & CH_2- & OH & & \end{array}$ Parent : Propane Name : Propan-1-ol	$\begin{array}{ccccccc} & 1 & 2 & 3 & 4 & & \\ & CH_3 & -CH= & CH- & CH_3 & & \end{array}$ Parent : Butane Name : But-2-ene	$\begin{array}{ccccccc} & 1 & 2 & 3 & 4 & & \\ & HC \equiv C & -CH_2- & CH_3 & & & \end{array}$ Parent : Butane Name : But-1-yne
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8. ISOMERISM

- Compounds having the same molecular formula but different structures are called **isomers**.
- Isomerism arises due to different arrangements of atoms.

Types of Isomerism

Structural (Constitutional) Isomerism

Compounds differ in the sequence in which atoms are joined.

Stereoisomerism

Compounds have same sequence but differ in spatial arrangement of atoms.

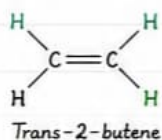
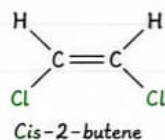
(A) Structural Isomerism

1. Chain Isomerism Different carbon skeleton. n-Butane $CH_3-CH_2-CH_2-CH_3$ (straight chain) Isobutane (2-methylpropane) $\begin{array}{c} CH_3-CH-CH_3 \\ \\ CH_3 \end{array}$ (branched chain)	2. Position Isomerism Functional group / double bond at different position. Propan-1-ol $CH_3-CH_2-CH_2-OH$ Propan-2-ol $CH_3-CH(OH)-CH_3$	3. Functional Group Isomerism Different functional groups. Ethanol (Alcohol) CH_3-CH_2-OH Dimethyl ether (Ether) CH_3-O-CH_3	4. Metamerism Different alkyl groups on either side of divalent group. Diethyl ether $CH_3CH_2-O-CH_2CH_3$ Methyl propyl ether $CH_3-O-CH_2CH_2CH_3$
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(B) Stereoisomerism Due to different spatial arrangement.

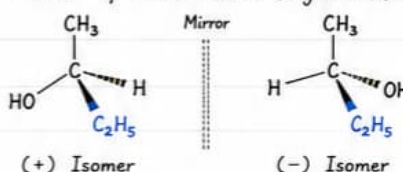
(i) Geometrical Isomerism (Cis-Trans)

Restricted rotation around $C=C$ bond.



(ii) Optical Isomerism

Presence of chiral carbon (asymmetric).



Isomerism is very important in living systems as small change in structure can change properties completely.

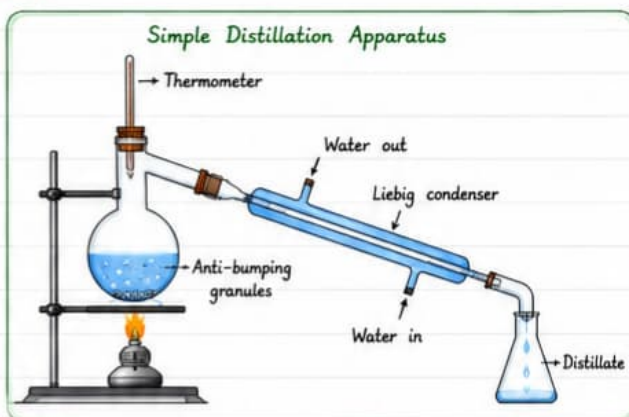




9. METHODS OF PURIFICATION OF ORGANIC COMPOUNDS

♦ Organic compounds are purified by the following methods:

- Sublimation**: Solid compound changes directly into vapour and on cooling changes back to solid.
Example: Naphthalene, camphor, iodine.
- Crystallisation**: Impure solid is dissolved in a suitable solvent and allowed to crystallise.
- Distillation**: Used for liquid compounds having sufficient difference in boiling points.
- Steam Distillation**: Used for water immiscible compounds which are volatile in steam.
- Differential Extraction**: Based on difference in solubility of a compound in two immiscible solvents.



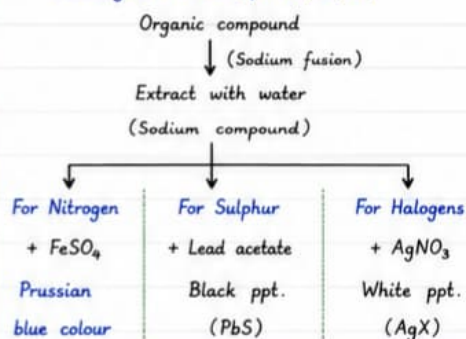
Use: To separate a volatile liquid from a non-volatile impurity or two miscible liquids with diff. b.p.

10. QUALITATIVE ANALYSIS OF ORGANIC COMPOUNDS

♦ It helps to detect the presence of different elements in an organic compound.

Element	Test	Observation (Positive Test)
Carbon (C)	Heating with CuO	CO ₂ evolved turns lime water milky
Hydrogen (H)	Heating with CuO	Water droplets on cool surface
Nitrogen (N)	Lassaigne's test	Prussian blue colour with FeSO ₄
Sulphur (S)	Lassaigne's test	Black ppt. of PbS with lead acetate
Halogens (X)	Beilstein test	Green flame (for halogens)
Oxygen (O)	Heating with Cu	Black CuO formed

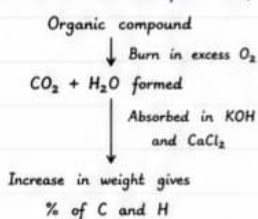
Lassaigne's Test (for N, S, X)



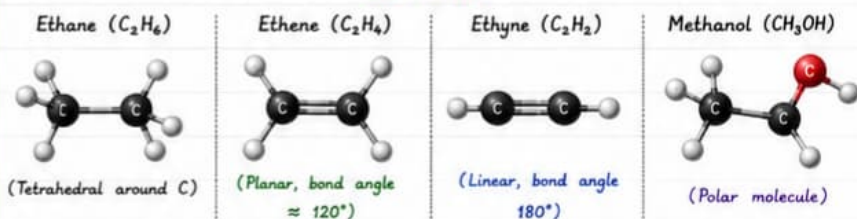
11. QUANTITATIVE ANALYSIS (ELEMENTAL ANALYSIS)

- It determines the percentage composition of elements (C, H, N, S, X) in an organic compound.
- Helps in finding out empirical and molecular formula.

Combustion Method (for C & H)



3D Models



12. CHARACTERISATION OF ORGANIC COMPOUNDS

- Characterisation means confirming the structure of an organic compound.
- It is done using physical and chemical methods.

Physical Methods

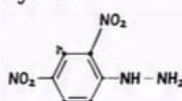
- ✓ Melting point (m.p.)
- ✓ Boiling point (b.p.)
- ✓ Solubility
- ✓ Refractive index
- ✓ Density
- ✓ Spectroscopic techniques

Chemical Methods

- ✓ Preparation of derivative (e.g., 2,4-DNP, semicarbazone)
- ✓ Functional group tests
- ✓ Oxidation
- ✓ Reduction
- ✓ Hydrolysis

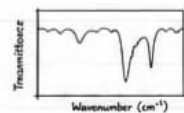
2,4-DNP Test

Aldehydes and ketones form yellow/orange ppt. with 2,4-dinitrophenylhydrazine.

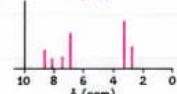


Spectroscopic Methods

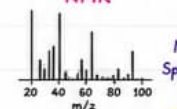
- ✓ IR Spectroscopy - identifies functional groups.
- ✓ NMR Spectroscopy - gives information about carbon skeleton and H environment.
- ✓ Mass Spectrometry - gives molecular mass and structure fragments.



IR



NMR



Mass Spectrum

★ **Key Point**: Organic chemistry is the chemistry of carbon compounds. Understanding its principles helps in exploring millions of compounds around us!





Organic Chemistry: Some Basic Principles and Techniques

13. IDENTIFICATION OF ORGANIC COMPOUNDS

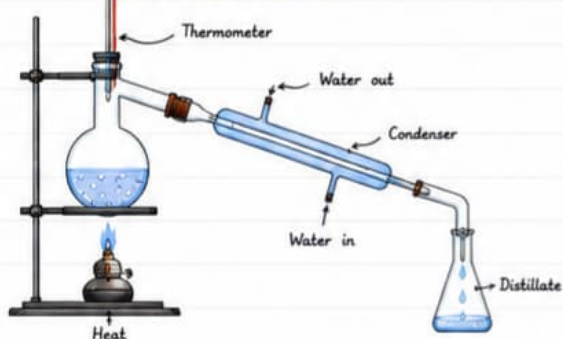
- Identification is done by chemical tests which indicate the presence of a functional group in the compound.

Functional Group	Test	Observation (Positive)
-OH (Alcohol)	Lucas test	Turbidity appears (immediate for 3°, slow for 2°)
-COOH (Acid)	NaHCO ₃ test	Brisk effervescence of CO ₂
-CHO (Aldehyde)	Tollens' test	Silver mirror is formed
C=O (Ketone)	2,4-DNP test	Orange / yellow ppt. forms
C=C (Alkene)	Bromine water test	Orange colour decolourises
C≡C (Alkyne)	Ammoniacal AgNO ₃ test	White ppt. of silver acetylide

16. DISTILLATION

- Technique used to separate a liquid from a solution or from a mixture based on difference in b.p.
- Types : Simple, Fractional and Steam distillation.

Simple Distillation Setup



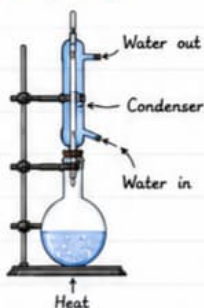
18. APPLICATIONS OF ORGANIC CHEMISTRY

	Medicines	→ Many drugs like paracetamol, aspirin, antibiotics are organic compounds.
	Polymers	→ Plastics, synthetic fibres, rubber and resins are organic in nature.
	Perfumes & Flavours	→ Esters and other organic compounds give pleasant smell and taste.
	Pesticides	→ Organic compounds are used to protect crops from insects and weeds.
	Fuels	→ Petrol, diesel, LPG, CNG are organic compounds.
	Dyes & Pigments	→ Used in textiles, paints, inks and food colouring.

14. ELEMENTARY IDEA ABOUT SOME LABORATORY TECHNIQUES

- These techniques are widely used for handling organic compounds in the lab.

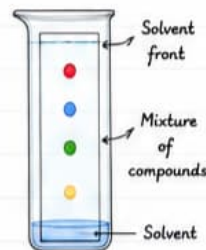
(i) Refluxing



(ii) Filtration



(iii) Chromatography

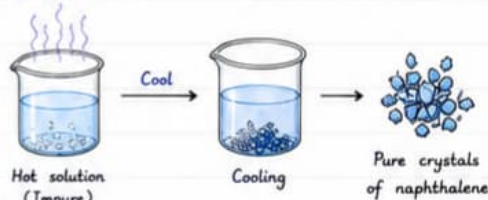


Purpose : To separate, purify and identify organic compounds using different physical and chemical properties.

15. CRYSTALLISATION

- Impure solid is dissolved in hot solvent.
- On cooling, pure crystals separate out.
- Used for purification of solid organic compounds.

Example : Purification of naphthalene



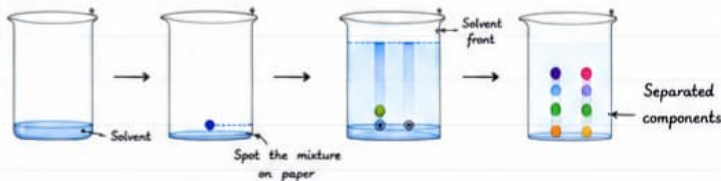
Key Tip

Choose a solvent in which compound is soluble in hot but sparingly soluble in cold.

17. CHROMATOGRAPHY

- Technique used to separate components of a mixture.
- Based on the difference in adsorption and partition.
- Types : Paper chromatography, Column chromatography, Thin layer chromatography (TLC).

Example : Separation of dyes by paper chromatography



19. SUMMARY - KEY POINTS

- ★ Organic chemistry is the chemistry of carbon compounds.
- ★ Carbon shows tetravalency and catenation.
- ★ Compounds are classified into homologous series.
- ★ Functional group determines the chemical properties.
- ★ Purification and qualitative/quantitative analysis help in studying organic compounds.
- ★ Laboratory techniques like distillation, crystallisation and chromatography are essential.
- ★ Organic chemistry has vast applications in daily life.

Quick Revision

- ✓ Remember major functional groups and their tests.
- ✓ Learn IUPAC rules for naming.
- ✓ Understand purification & identification techniques well.

- ✓ Practice numericals of elemental analysis.
- ✓ Relate concepts with real life examples.

Concepts clear today → Success tomorrow!

Keep Practising
Keep Growing! 😊